16.4. The TD reaction is used to produce 14 MeV neutrons, which are considered to be emitted isotropically from the target. What is the fast neutron "flux" at 5 cm from the target when the ion current is 0.2 mA and the acceleration voltage is 300 kV? Use Figure 16.11.

16.5. Protons are accelerated to 12 GeV in a synchrotron in which the bending magnets have a maximum field strength of 14.3 T. What is the radius of curvature of the proton orbit?

16.6. In a linear accelerator for protons the first drift tube is 1 cm long and the accelerating potential is 25 kV. (a) The speed of light is not approached in the first drift tube. How long should the second and third drift tubes be? (b) What should the full-wave frequency of acceleration be?

16.7. Calculate the maximum energy (a) for protons, deuterons, and helium ions in a cyclotron, whose maximum orbit diameter is 1.25 m and whose frequency is 12 MHz. (b) What magnetic field strength would be required in each case?

16.8. A thin Au target is irradiated with a beam of 800 MeV $^{16}O^{2+}$ ions during 2 hours. After the irradiaten a faraday cup (with current integrator) behind the target shows a total accumulated charge of 92.3 μC. What was the average beam intensity (oxygen ions per second)? Consider the charge-stripping effect of the target.

16.9. In a VdG accelerator a 100 μA beam of He^{2+} ions is accelerated to an energy of 5 MeV before striking a target. How many grams of radium are required to provide the same number of α-particles?

16.10. A cyclotron can accelerate $^{4}He^{2+}$-ions to 35 MeV. (a) What is its *K*-value? To what energy would it accelerate (b) $^{16}O^{6+}$ and (c) $^{18}O^{8+}$ ions?

16.11. LITERATURE

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CHAPTER 17

Production of Radionuclides

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This chapter discusses production of radionuclides for beneficial use in science, medicine and technology. The nuclear fundamentals for the production processes have been given in Chapters 6, 10, 11 and 16. The formation of radionuclides is discussed in several Chapters: e.g. cosmogenic reactions leading to the formation of short-lived radionuclides in nature (Ch.12 and 13); thermonuclear reactions leading to the formation of long-lived radioactivity in the universe (Ch. 12); the synthesis of trans-uranium elements (Ch. 14 and 19-21). The production and isolation of separated fission products is treated separately (Ch. 19—21). This chapter discusses aspects of fundamental importance to the production of radionuclides by a variety of methods. Initially the principles are reviewed

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